SAHIL DUWAL

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SUMMARY

Highly motivated Computer Engineer with significant experience in AI, Machine Learning, and Web Development. Proficient in designing intelligent deep learning architectures, computer vision algorithms, and resilient full-stack web applications with payment and cloud integrations. Passionate about applying cutting-edge AI technology to build efficient, impactful digital products.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C, C++, HTML, CSS

AI, Machine Learning & Vision: PyTorch, Computer Vision, Deep Learning, Natural Language Processing (NLP), scikit-learn

Full-Stack Web Development: Node.js, NestJS, React, Next.js, FastAPI, Tailwind CSS, RESTful APIs, Stripe, Prisma

Data Science & Databases: Pandas, NumPy, Matplotlib, MySQL, PostgreSQL, Data Analysis, Statistical Modeling

Developer Tools & Frameworks: Git, Linux/Unix, Git Workflow, AWS (S3, ECS, Cognito), Terraform, GitHub Actions, Jest, Playwright, LaTeX

SOFT SKILLS

Technical Communication • Collaborative Problem Solving • Analytical Thinking • Adaptability • Project Management

EXPERIENCE

CVArcher

Full Stack Developer *August 2026 – November 2026*

- Built Stripe Connect marketplace payments (embedded onboarding, escrow checkout, scheduled payouts, refunds, disputes) with idempotent webhooks.
- Designed a 20+ state booking state machine; fixed six payout-path defects, verified via Stripe test transfers.
- Implemented Google Calendar two-way sync (OAuth, push webhooks, iCal) and S3 storage with STS and signed URLs.
- Hardened a NestJS API (CSRF/brute-force guards, ownership checks on 15 endpoints) and remediated a 17-finding P0 audit; API tests grew 127 → 383.

Full Stack Development Trainee *May 2026 – July 2026*

- Built Next.js/NestJS features in TypeScript and integrated Google OAuth and Stripe (checkout, webhooks, payouts).
- Executed 150+ QA test cases and logged defects, including a missing role check that exposed client routes to MCs.

Next Wave AI

Data Annotation Intern *October 2025 – December 2025*

- Annotated high-quality image datasets (including objects and media covers) to support the training and development of computer vision models.
- Generated structured JSON schemas for AMS360 to facilitate seamless workflow automation.
- Performed comprehensive website reviews and functional testing to identify UI/UX bottlenecks and performance issues.
- Evaluated Large Language Model (LLM) security behaviors and vulnerabilities through targeted prompt testing and adversarial analysis.
- Researched and conducted hands-on capability assessments of emerging AI video generation tools.

Code for Nepal

DataCamp Fellowship *2025 – 2026*

- Selected for an intensive data science fellowship program focused on advanced analytical methodologies.
- Accelerated technical expertise in Python, exploratory data analysis (EDA), and practical machine learning applications.

Khwopa College of Engineering

Tech Support Member – Hult Prize On-Campus Program *2024 – 2025*

- Delivered comprehensive technical support for event management, complex presentation setups, and digital logistics.
- Assisted participating teams in troubleshooting IT-related and software issues during high-stakes workshops, mentoring sessions, and hackathons.

- Ensured flawless technical operations across multiple campus-wide events, directly contributing to the program's operational success.

EDUCATION

Bachelor of Computer Engineering *2022 – 2026*

Khwopa College of Engineering, IOE – Libali, Bhaktapur

- Pursued a comprehensive program emphasizing Artificial Intelligence, Machine Learning, and Software Engineering.
- Engaged in practical projects and research focused on intelligent systems and real-world problem solving.

CERTIFICATES & COURSES

- Learn HTML – Programiz Pro | 2024
- SQL Basics – Programiz Pro | 2024
- Learn Python Basics – Programiz Pro | 2024
- Python Essentials I – Cisco Networking Academy | 2024
- Introduction to Cybersecurity – Cisco Networking Academy | 2024
- Introduction to IoT – Cisco Networking Academy | 2024
- AI Fundamentals – DataCamp | 2025
- Data Skills for Business – DataCamp | 2025
- Understanding Data Topics – DataCamp | 2025
- EU AI Act Fundamentals – DataCamp | 2025

ACHIEVEMENTS

Hult Prize Nepal *2022 – 2023*

- Reached semi-finals with a campus-team startup idea, presenting an innovative social impact solution.

CodeYatra Hackathon *2025*

- Participated in a national-level innovation hackathon focused on AI-driven problem-solving.

Locus Pattern Verse *2025*

- Advanced to Phase 2 in a national tech competition with an AI-powered rug-design generation system.

PROJECTS

AI, Machine Learning & Data Science

Deepfake Detection & Localization
Developed a hybrid ViGNN–U-Net model for detecting and localizing deepfake content with high accuracy and robustness across various manipulation types.
Tech: *PyTorch, Computer Vision, Deep Learning, U-Net, ViGNN*

Quantum-Enhanced BERT Model — View Repository
Hybrid classical-quantum ML pipeline combining multilingual BERT embeddings with variational quantum circuits (VQCs) for low-resource Nepali offensive-speech detection.
Tech: *PyTorch, BERT, Variational Quantum Circuits, NLP*

Shruti: AVSR in Nepali Language — View Repository
Audio-visual speech recognition (AVSR) system built for the Nepali language, combining acoustic and visual cues for improved recognition accuracy.
Tech: *Deep Learning, Computer Vision, NLP, Python*

Ghumti: AI Travel Assistant
Built an intelligent chatbot powered by Llama 3 to provide real-time local bus routes, schedules, and optimized travel information.
Tech: *Llama 3, NLP, Chatbot, API Integration*

PDF Q&A RAG Bot — View Repository
Local retrieval-augmented generation (RAG) application enabling semantic Q&A across multiple PDFs, running entirely offline.
Tech: *Llama2, LangChain, DocArray, Sentence-Transformers*

Captioning Image — View Repository
Image captioning model combining a pretrained ResNet50 encoder with an LSTM decoder to generate descriptive captions for input images.
Tech: *PyTorch, ResNet50, LSTM*

Web & Software Development

CentraSpend – Expense Tracker — View Repository
Full-stack personal expense tracking application with account management and transaction history.
Tech: *React, Express, Node.js, MongoDB*

RecipeBook: Cooking Recipe Portal — View Repository
Web application for storing, organizing, and managing cooking recipes with a full client-server architecture.
Tech: *React, FastAPI, MySQL*

Personal Portfolio Website — Portfolio 2.0
Personal portfolio website showcasing skills, experience, and completed projects; rebuilt as a v2 with an updated design.
Tech: *TypeScript, React, CSS*